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URL of this page: https://medlineplus.gov/lab-tests/basic-metabolic-panel-bmp/

Basic Metabolic Panel (BMP)

What is a basic metabolic panel (BMP)?

A basic metabolic panel (BMP) measures eight different substances in your blood. It provides important information about your body's fluid balance [https://medlineplus.gov/fluidandelectrolytebalance.html] , your metabolism (the process your body uses to make energy from food you eat), and how well your kidneys are working. This test is often used to check your general health.

A BMP test includes measurements of the following:

- Glucose [https://medlineplus.gov/bloodglucose.html] , a type of sugar and your body's main energy source. If your glucose level is high, it may be a sign of diabetes [https://medlineplus.gov/diabetes.html] .
- Calcium [https://medlineplus.gov/calcium.html] , one of the body's most important minerals. Calcium is essential for the proper functioning of your nerves, muscles, and heart. It also helps with blood clotting, an important process that prevents you from losing too much blood when you get injured.
- Sodium [https://medlineplus.gov/lab-tests/sodium-blood-test/] , potassium [https://medlineplus.gov/lab-tests/potassium-blood-test/] , carbon dioxide [https://medlineplus.gov/lab-tests/carbon-dioxide-co2-in-blood/] , and chloride [https://medlineplus.gov/lab-tests/chloride-blood-test/] . These are electrolytes, which are electrically charged minerals [https://medlineplus.gov/minerals.html] that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. They help move nutrients into your cells and waste out. They also help your nerves, muscles, heart, and brain work well.
- BUN [https://medlineplus.gov/lab-tests/bun-blood-urea-nitrogen/] **(blood urea nitrogen)** and creatinine [https://medlineplus.gov/lab-tests/creatinine-test/] , waste products removed from your blood by your kidneys.

Abnormal levels of any of these substances or a combination of them can be a sign of a health problem.

Other names: BMP, chemistry panel, chemistry screen, chem 7, electrolyte panel

What is it used for?

A BMP can be used to screen or diagnose health conditions or monitor treatment. This test may also be used to check how well your body functions. It can provide information on your:

- Kidney function

- Fluid and electrolyte balance
- Blood glucose [https://medlineplus.gov/bloodglucose.html] levels
- Acid and base balance
- Metabolism

Why do I need a BMP?

A BMP is often done as part of a regular checkup [https://medlineplus.gov/healthcheckup.html] . This may help your health care provider find health problems before you have any symptoms.

You may also need this test if you:

- Have general symptoms such as fatigue [https://medlineplus.gov/fatigue.html] , confusion, breathing problems [https://medlineplus.gov/breathingproblems.html] , or vomiting [https://medlineplus.gov/nauseaandvomiting.html] that lasts a long time
- Are being treated in the emergency room
- Are being monitored for a certain chronic (long-lasting) condition, such as high blood pressure [https://medlineplus.gov/highbloodpressure.html] or kidney disease [https://medlineplus.gov/kidneydiseases.html]

What happens during a BMP?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You may need to fast [https://medlineplus.gov/lab-tests/fasting-for-a-blood-test/] (not eat or drink) for eight hours before the test.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

If any one result or combination of BMP results were not normal, it could be a sign of several different health conditions. These include kidney disease, breathing problems , and complications from diabetes [https://medlineplus.gov/diabetescomplications.html] . You will likely need more tests to confirm or rule out a specific diagnosis.

To understand what your test results may mean, your provider may also consider your symptoms, your medical history, and the results of other tests you had. If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results [https://medlineplus.gov/understandingmedicalresearch.html] .

Is there anything else I need to know about a BMP?

There is a similar test to a BMP called a comprehensive metabolic panel [https://medlineplus.gov/lab-tests/comprehensive-metabolic-panel-cmp/] (CMP). A CMP includes the same eight tests as a BMP, plus six more tests, which measure certain proteins and liver enzymes. The additional tests are:

- Albumin [https://medlineplus.gov/lab-tests/albumin-blood-test/] , a protein made in the liver
- Total protein, which measures the total amount of protein in the blood
- ALP [https://medlineplus.gov/lab-tests/alkaline-phosphatase/] (alkaline phosphatase), ALT [https://medlineplus.gov/lab-tests/alt-blood-test/] (alanine transaminase), and AST [https://medlineplus.gov/lab-tests/ast-test/] (aspartate aminotransferase). These are different enzymes (proteins that speed up certain chemical reactions in your body) made by the liver.
- Bilirubin [https://medlineplus.gov/lab-tests/bilirubin-blood-test/] , a waste product made by the liver

Your provider may order a CMP instead of a BMP to get a more complete picture of the health of your organs or to check for liver disease or other specific conditions.

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